

Required fields (official storyboard cover)

1. Storyboard Title	FloodResilience ASEAN — Data-Driven Urban Flood Risk Intelligence for Yangon
2. Team Name	Zenith
3. Institution	University of Information Technology (UIT)
4. Country Represented	Myanmar
5. SDG(s)	SDG 11 (11.5, 11.b) · SDG 13 (13.1, 13.3)
6. Brief Description	Using 45 years of satellite rainfall, census and critical-infrastructure data in SAP Analytics Cloud, we map flood risk for all 45 Yangon townships — identifying the 9 highest-risk townships where 2.6 million people live — and deliver a decision-support tool for cities across ASEAN.

About Us

Thar Lun Myint Myat Tun · Team Leader

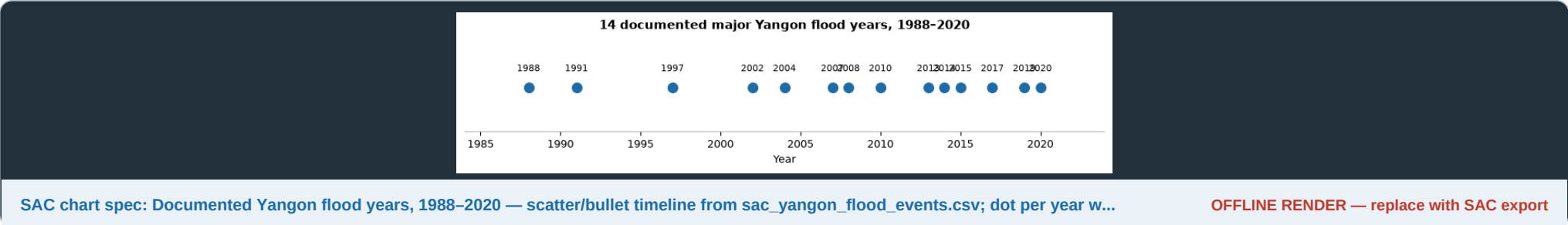
La Yaung Htut · Team Member

University of Information Technology (UIT)

When the monsoon arrives, who in Yangon is at risk?

Floods are a documented, recurring reality — not a hypothetical. Yangon Region has recorded major flood years repeatedly since 1988, and Myanmar is among ASEAN's most flood-affected countries.

- 14 documented major flood years in Yangon Region since 1988 — 1988, 1991, 1997, 2002, 2004, 2007, 2008, 2010, 2013, 2014, 2015, 2017, 2019, 2020.
- 33 DFO-documented flood events across Myanmar (1990–2023).
- Monsoon rains are extreme and seasonal: peak month July averages 698.9 mm.
- Cities concentrate exposure: ~7.4 million people live across Yangon's 45 townships.



14

documented flood years (Yangon)

33

DFO events (Myanmar)

698.9 mm

July mean peak rainfall

TAKEAWAY Flooding in Yangon is frequent, recent and measurable — the only question is where it will hurt most.

Sources: PIAHS 2024; Sritarapipat 2017; OCHA 2017; UNOSAT 2020; DFO; CHIRPS v2.0 1981–2026; 2014 Myanmar Census (via MIMU).

Not a local data exercise — an ASEAN priority.

This project is anchored to international and regional commitments, which is what makes it relevant and fundable beyond Yangon.

Framework / element	Target / element	How FloodResilience ASEAN supports it
SDG 11.5	Reduce deaths & people affected by disasters	Township-level risk ranking enables pre-season targeting
SDG 11.b	Adopt DRR strategies; build resilient cities	Risk model + open web tool = replicable DRR method
SDG 13.1	Resilience to climate-related hazards	Rainfall (1981–2026) drives the hazard layer
SDG 13.3	Build climate capacity	Method & data fully documented; anyone can replicate
ASCC B.2	Equitable access for all	Free, open web tool; no paywall for townships
ASCC D.4	Protect vulnerable groups in climate crises	Exposure layer flags elderly/children (census age structure)
AADMER	ASEAN disaster management	Outputs align to national/regional disaster data formats

TAKEAWAY The alignment table answers 'Relevancy & Impact' — 20% of the judging score.

Sources: UN SDG targets 11.5/11.b/13.1/13.3; ASEAN ASCC Blueprint 2025 (B.2, D.4); AADMER (2013).

One question, one region, one framework.

Research question: “Which Yangon townships face the highest flood risk — and which populations, schools and health facilities are exposed there?”

Approach: open data → hazard · exposure · vulnerability → risk ranking → decision support. The full pipeline runs offline in Python, then imports into SAP Analytics Cloud for exploration and presentation.

Yangon Region — 45 townships by district



SAC chart spec: Yangon Region overview — 45 townships from geoBoundaries + MIMU P-codes (MMR013001–MMR013045), districts East...

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Honesty box — what this is, and is not:

- We compute relative, township-level risk — not a flood forecast.
- All data is open and cited; every township row is traceable to its source.

TAKEAWAY Clarity of scope = credibility.

Sources: geoBoundaries; MIMU township P-codes; project pipeline (floodresilience package).

Open data, openly cited. 547 months of rainfall. 7.4 million people.

Every number on every slide traces to an open, cited source — that is the defense for 'Analysis & Insights' and 'Viability'.

Layer	Dataset	Key figure
Rainfall	CHIRPS v2.0 (UCSB)	547 monthly observations, 1981-01..2026-07
Hazard inputs	World Bank / GFDRR rainfall indices	2022+, 10-day scale, district level
Elevation	Copernicus DEM 30 m	mean elevation -9.2 m (Hlaingtharya) .. 37.5 m (Taikkyi)
Flood events	DFO Flood Archive + PIAHS/UNOSAT/OCHA	14 Yangon years; 33 Myanmar events
Population	2014 Myanmar Census (via MIMU/ODM)	7,360,703 total; urban/rural; ages 0–14 & 65+
Facilities	OSM / HDX education & health polygons	1,320 schools; 1,178 health facilities
Boundaries	geoBoundaries + MIMU	45 townships, P-codes matched

547

rainfall months

45

townships mapped

1,320

schools located

1,178

health facilities

Sources: full catalog in [data/source_catalog.csv](#) — every dataset open and licensed for reuse.

Risk = 0.4·Hazard + 0.35·Exposure + 0.25·Vulnerability — weighted by the data.

- Weights from the WB/GFDRR risk literature, applied to township-level normalized indicators; all components are documented.
- Quantile classing produces 5 risk classes (1 = lowest .. 5 = highest).
- Robustness: $\pm 10\%$ sensitivity re-simulation changes a township's class by only 4.04 ranks on average — the ranking is stable, not noise.
- Hazard justification: an extreme-rainfall month classifier (Random Forest) scores F1 = 0.667, ROC-AUC = 0.949 — rainfall signals genuinely predict wet-season extremes.

Risk = 0.4·Hazard + 0.35·Exposure + 0.25·Vulnerability



SAC chart spec: Risk formula decomposition — donut/waterfall: 0.40 Hazard · 0.35 Exposure · 0.25 Vulnerability (calculated measure on s...

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0.4 / 0.35 / 0.25

Hazard / Exposure / Vulnerability weights

4.04

avg rank shift under $\pm 10\%$ sensitivity

0.949

ROC-AUC extreme-rainfall classifier

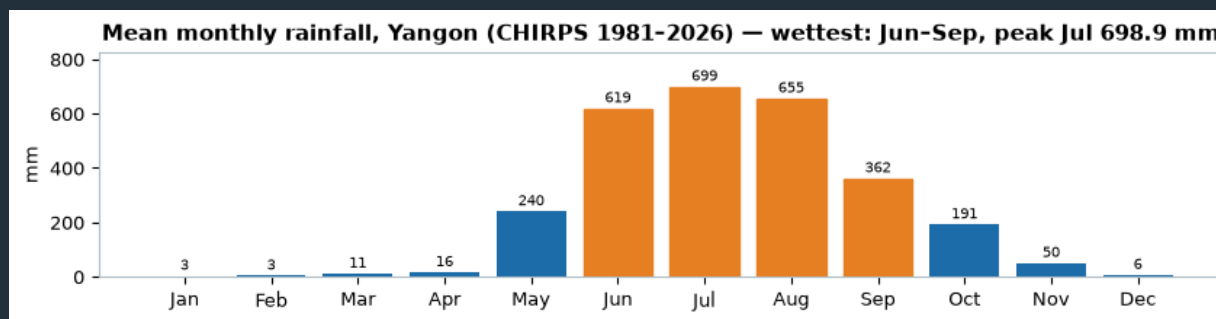
Honesty: weights are transparent and tunable; the full sensitivity table ships in the appendix (risk_sensitivity.csv).

TAKEAWAY A model the judges can interrogate beats a black box.

Sources: WB/GFDRR risk methodology; floodresilience analysis pipeline; outputs/tables/risk_sensitivity.csv.

The monsoon is not ‘a season’ — it is a peak that must be planned for.

- Peak month July averages 698.9 mm (CHIRPS 1981–2026); wettest months are July, August, June, September.
- Documented flood years averaged 2,903 mm/yr vs a 2,844 mm/yr long-run mean — flood years are measurably wetter.
- Long-run trend is small (+15 mm/yr, 1981–2025); interannual variability dominates — we do not overclaim a climate trend.



SAC chart spec: Monthly rainfall climatology — bar + line over 12 months; calculated measure for wet-season share; Rank Top-N = 6 for w...

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698.9 mm

July mean peak

+59 mm

flood-year rainfall anomaly

Jun–Sep

the 4 wettest months

TAKEAWAY Seasonality tells us when; the risk model tells us where.

Sources: CHIRPS v2.0 monthly, bbox mean; flood years per PIAHS 2024, Sritarapipat 2017, OCHA 2017, UNOSAT 2020.

Five townships you should know before the next monsoon.

- Highest risk (class 5): Kayan 58.3 · Thongwa 57.4 · North Okkalapa 57.2 · Thanlyin 53.2 · Hlaingtharya 52.5 (risk index 0–100).
- Low elevation aligns with high risk: the lowest-lying townships sit along the Yangon, Bago and Hlaing rivers and the delta fringe (mean elevation –9.2 m .. 37.5 m).

Township flood risk (risk_100) — dark red = highest



CHART 1: SAC chart spec: Choropleth risk map — risk_100 on all 45 townships (geoBoundaries + MIMU), risk ramp 1-green..5-red. THE h...

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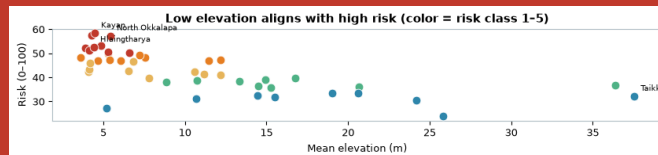


CHART 2: SAC chart spec: Elevation vs risk_100 scatter — x = elevation, color = risk class — shows the lowland-hazard alignment. Add In...

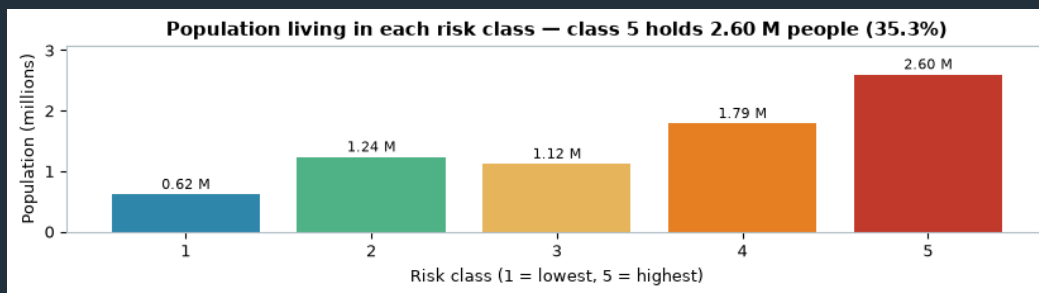
OFFLINE RENDER — replace with SAC export

TAKEAWAY Geography is the story: the delta fringe is where risk concentrates.

Sources: Copernicus DEM 30 m; CHIRPS; 2014 Census; geoBoundaries/MIMU boundaries.

Risk is not abstract — it is 2.6 million people, 1,320 schools, 1,178 clinics.

- 2,599,370 people (35.3% of the region) live in the highest-risk townships (class 5).
- District populations: East 2,366,659 · North 2,606,670 · South 1,417,724 · West 969,650 (total 7,360,703; urban + rural = total everywhere).
- Schools and health facilities exist in every township — no flood event is free of public-service impact.
- Age vulnerability: children (0–14) and elderly (65+) enter the vulnerability sub-index (2014 Census age structure).



SAC chart spec: Population at risk by risk class — 5 bars; highlight class 5 (2.6 M). Bubble map: schools & health facilities sized/colored ...

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2.60 M

people in class-5 townships

35.3%

of Yangon's population

2,498

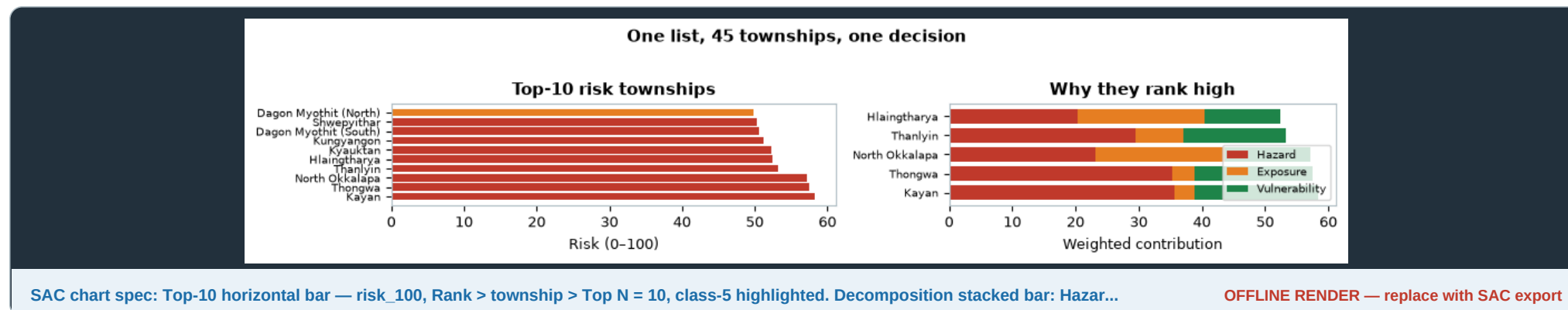
facilities mapped

TAKEAWAY We count people and services, not just polygons.

Sources: 2014 Myanmar Census (DoP via MIMU); OSM/HDX facility polygons; geoBoundaries.

From 45 townships to one prioritized list.

Ranking every township on the same scale turns raw data into a decision: which townships get early-warning, evacuation drills and drainage investment first.

**58.3**

top risk score (Kayan)

9

townships in class 5

4.04

avg rank shift under sensitivity

Optional: run SAC Smart Discovery (target: risk_100, entity: township) and caption the key-influencer panel as “SAC Smart Discovery”.

TAKEAWAY A prioritized, explainable list — that is what decision-makers asked for.

Sources: outputs/tables/risk_scores.csv; floodresilience risk_index analysis.

Four evidence-backed answers to ‘so what?’

1. Flood years are wet years.

Flood years averaged 2,903 vs 2,844 mm/yr. • Limitation: correlation $\neq$ causation; rivers/tides/drainage also matter.

2. Risk follows elevation and the rivers.

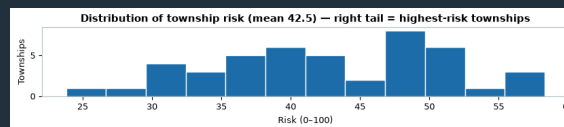
Lowest-lying townships dominate class 5. • Limitation: DEM is a surface model; no flood depth.

3. Exposure is concentrated; services are everywhere.

35.3% of people in class 5; schools/clinics in all 45. • Limitation: OSM may undercount informal facilities.

4. The ranking is robust.

Sensitivity shifts rank by only 4.04. • Limitation: $\pm 10\%$ band tested; wider bands are future work.



SAC chart spec: SAC Smart Discovery key-influencer panel (captured screenshot) or risk-distribution histogram.

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TAKEAWAY Insights are evidence-backed and limitation-aware — exactly what 'Analysis & Insights' rewards.

Sources: CHIRPS; Copernicus DEM; 2014 Census; OSM/HDX; outputs/reports/key_insights.md.

From analysis to action: FloodResilience ASEAN.

A decision-support platform (open web prototype) that turns the SAC risk model into township-level answers anyone can use.

- Live risk explorer — all 45 townships, ranked and filterable by district and risk class.
- Scenario engine — raise/lower rainfall, exposure and vulnerability and see rankings change live.
- Critical-facility locator — schools and clinics against township risk.
- Data dictionary with citations — every field traceable.



Prototype: open web build (SvelteKit); screenshots must be captioned 'prototype'. The risk model itself is built and exported in SAC.

TAKEAWAY The model only matters if it changes a decision — the tool makes the decision visible.

Sources: outputs from the floodresilience pipeline (src/lib/data/*.json); SAC risk model.

Who does what, and when.

Phase	Window	Milestones
Phase 1 — Pilot	0–6 months	Validate with Yangon Region DRR office; 5 townships get risk briefs
Phase 2 — Scale	6–18 months	Web tool public; schools & clinics onboarding; monsoon early-warning link
Phase 3 — ASEAN	18–36 months	Replicate pipeline for a 2nd city; AADMER/ASCC reporting alignment

Stakeholders:

- Region DRR office (validates) · Township administrators (decide) · MIMU/HDX (data custodians)
- Schools & clinics (services at risk) · Researchers (open method) · SAP Analytics Cloud (analytics platform)

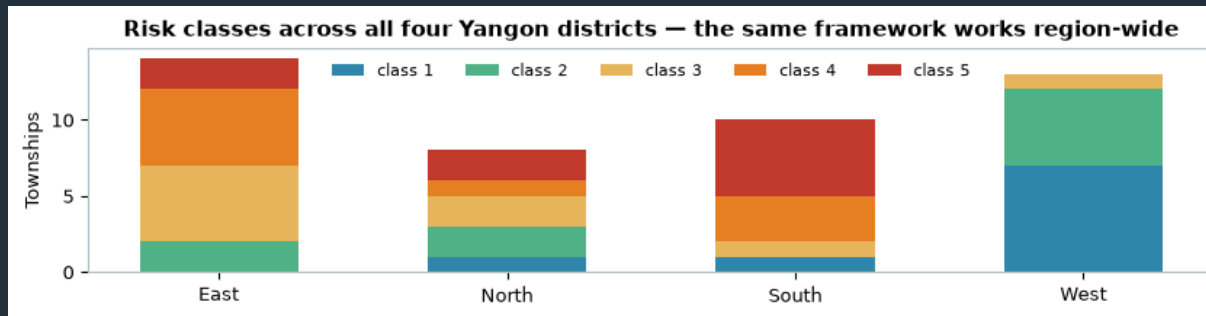
Honesty note: cost/revenue claims are NOT fabricated — pilot costs depend on the host government arrangement (no invented budgets).

TAKEAWAY A plan the user can actually run — no invented budgets.

Sources: project roadmap (docs/sac_storyboard.md); methodology & viability sections.

Built for Yangon. Designed for ASEAN.

- Common (reusable) layers: CHIRPS/GPM rainfall, Copernicus DEM, census, OSM facilities, DFO — these exist for every ASEAN country.
- Local (swap-in) layers: admin boundaries + P-codes, national census, flood-event catalogues.
- Replication path: fetch open layers for target city → run the documented pipeline → import to SAC and publish the same web tool. Low marginal cost = high viability.



SAC chart spec: ASEAN data availability — bar counting the 5 core layers available per country, or a mini-map of pilot candidate cities.

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5

core open layers, reusable

3

steps to replicate

0

proprietary data required

TAKEAWAY Open data + documented method + cloud analytics = repeatable anywhere.

Sources: data/source_catalog.csv; asean_scalability.md.

The next monsoon is coming. We now know where it will hurt — and what to do about it.

- 45 townships ranked on one transparent risk scale; 9 identified as highest risk.
- 2.6 million people and 2,498 schools & clinics lie in the highest-risk zone.
- The ranking is robust ($\pm 10\%$ sensitivity $\rightarrow$ 4.04 rank shift) and every figure is open and cited.

Call to action: prioritize early-warning and infrastructure for Kayan, Thongwa, North Okkalapa, Thanlyin and Hlaingtharya first — then replicate the pipeline across ASEAN.

The 9 highest-risk townships (class 5) to prioritize



SAC chart spec: Closing visual — full risk map with class-5 townships highlighted, or the population-at-risk by-class summary.

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TAKEAWAY Team Zenith · University of Information Technology · Myanmar — data, method and tool are open and cited.

Sources: all figures verified in outputs/tables/ and outputs/reports/ — see References page.

References page — excluded from the 15-page limit.

- UCSB Climate Hazards Center. (1981–2026). CHIRPS v2.0 precipitation. <https://www.chc.ucsb.edu/data/chirps>
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- PIAHS (2024) Yangon flood events; Sritarapipat et al. (2017); OCHA (2017); UNOSAT (2020).
- ASEAN Secretariat. (2015). ASEAN Socio-Cultural Community Blueprint 2025. <https://asean.org/>
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Build tooling: SAP Analytics Cloud (Classic Design Experience); open Python pipeline (FloodResilience ASEAN) documented in the project repository.

All figures in this storyboard are computed from the cited open datasets — no values are invented.